CodeFest 2026 Project Proposal

Project Title:

FocusGuard – A Privacy-First Context-Aware AI Companion for Focus & Project Continuity

Team Name: Team MindMesh

Theme: SDG 3 – Good Health & Well-Being

FocusGuard promotes healthier digital work habits by helping users maintain focus, reduce executive overload, and recover from distractions without relying on restrictive productivity techniques.

Problem Statement

Students, developers, startup founders, researchers, and knowledge workers constantly switch between browsers, IDEs, AI tools, and documentation, often losing both focus and work context. For individuals with ADHD, these interruptions increase executive overload, making it difficult to resume tasks. Existing productivity tools rely on blockers or timers that ignore user intent and disrupt natural workflows. The real challenge is not avoiding interruptions—it's quickly recovering context and continuing meaningful work.

Proposed Solution

FocusGuard is a privacy-first, context-aware AI companion that helps users stay focused by understanding their work context instead of enforcing rigid productivity rules. Users begin each session by declaring a goal, and FocusGuard monitors lightweight browser and development activity to determine whether their actions align with that objective. Rather than treating inactivity as distraction, it waits for a configurable grace period before providing gentle, context-aware reminders. For developers, the VS Code extension and CLI maintain a **Persistent Project Context** in an encrypted local database, enabling seamless project resumption. Following a **Privacy-by-Design** architecture, raw

activity is converted into compact summaries and automatically deleted, with optional encrypted cloud backup for cross-device continuity.

Key Features

1. 🧠 **Context-Aware Focus Sessions:** AI evaluates whether current activities align with the user's declared goal instead of simply tracking application usage.
2. 🌿 **Smart Pause Detection:** Rather than interrupting immediately, FocusGuard waits for a configurable grace period before gently reminding users where they left off, distinguishing intentional breaks from unintentional distraction.
3. 🔒 **Privacy-First Session Compaction:** Temporary browser and editor events are summarised into contextual insights and automatically deleted after each session, minimising storage while protecting user privacy.
4. 🌐 **Browser Context Engine:** Monitors lightweight events such as tab lifecycle, active duration, focus changes, and context switching using privacy-filtered metadata to understand work context without storing browsing history.
5. 📁 **Persistent Project Context:** A CLI and VS Code extension maintain project objectives, architecture, milestones, AI summaries, pending tasks, and suggested next actions in an encrypted local database.
6. 📊 **Unified Dashboard:** Displays active projects, resume suggestions, focus analytics, productivity trends, and context-aware notifications through a single interface.
7. ⚙️ **Fully Customisable Experience:** Users can configure reminder intervals, intervention style, AI provider, focus sensitivity, quiet hours, local-only mode, and encrypted backup preferences.

Expected Social Impact

FocusGuard promotes healthier digital work habits by reducing cognitive overload through context-aware assistance rather than restrictive productivity methods. It helps students with ADHD regain focus, enables developers and researchers to seamlessly resume interrupted projects, and improves long-term productivity. Built

on a privacy-first architecture, it delivers intelligent support without excessive behavioural tracking, making it a scalable solution for individuals, educational institutions, software teams, and enterprises.

Technology Stack

Frontend: React, TypeScript, Tailwind CSS, Electron

Browser Extension: Chrome Extension API (Manifest V3), TypeScript

Developer Companion: VS Code Extension API, Node.js

Backend: FastAPI, REST API, WebSocket

AI Support: Flexible (DeepSeek, OpenAI, Claude, Gemini, Ollama (Local LLM))

Database: Encrypted SQLite, PostgreSQL (Optional Cloud Sync)

DevOps: Docker, GitHub Actions, Git

Innovation & Competitive Advantage

Unlike traditional productivity tools that rely on timers or website blockers, FocusGuard understands why users lose focus instead of simply reacting to it. Its combination of Smart Pause Detection, Persistent Project Context, and Privacy-First Session Compaction enables users to recover naturally without sacrificing privacy.

By combining context-aware AI, local-first processing, and user-controlled personalisation, FocusGuard transforms productivity software from a restrictive productivity monitor into an intelligent digital companion that supports healthier attention management while respecting user autonomy.